

Geometric Visualization of Two-Dimensional Spinors in $\mathbb{R}^3$

Alessandro Rigazzi

Switzerland

(*alrigazzi@gmail.com)

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Abstract

We introduce *hyperchords*, a geometric visualization for two-dimensional spinors (qubit states) embedded in $\mathbb{R}^3$. Whereas the Bloch-sphere map identifies states that differ by a global $U(1)$ phase, hyperchords make both the *relative* phase between spinor components and a *common* (fiber) phase directly readable from geometry, and render the spinor’s characteristic 4π periodicity as a visible double cover. The representation consists of two offset circular sections on a sphere (the *hyperchord faces*): their radii encode the component moduli, while face markers encode relative and common phase via rigid rotations. We derive the construction from the $SU(2)$ formalism, relate it to the Bloch vector and the Hopf fibration, and give an explicit forward and inverse mapping together with a complete rendering pipeline. We demonstrate static and animated views, including 2D projections that preserve the Born-rule interpretation of the component weights. An open-source Python reference implementation with interactive Jupyter notebooks is available at https://github.com/al-rigazzi/spinor_viz.

I. INTRODUCTION

Spinors were discovered by Élie Cartan [1] while he was working on the linear representation of simple groups. They are the most basic representation of the rotation and Lorentz groups and all other representations can be obtained as tensor products of spinors [2]. For this reason, they are often loosely described as the square roots of Euclidean (or Minkowskian) vectors [3]. Spinors are difficult to picture not only because their components are in general complex, but also because when the Euclidean space is rotated by an angle of 2π , they do not return to their initial state (like vectors do), but acquire a multiplicative phase of -1 . Only rotations of multiples of 4π are thus equivalent to the identity operator on spinors.

The standard approach for visualizing the state of a two-level quantum system is the Bloch sphere representation. While powerful and widely used, the Bloch sphere has a fundamental limitation: it represents only the *direction* associated with a spinor (via the Bloch vector $\mathbf{v}$), collapsing the global phase information. Furthermore, the Bloch sphere does not make explicit how the individual spinor components contribute to the overall state.

Spinors are not only intimately related to several group-theoretical and geometrical objects, such as quaternions, Hopf fibrations, stereographic projections, and Möbius transformations [4–6], but also of great importance for physics, as they are needed to correctly describe the wavefunction of spin- $\frac{1}{2}$ particles [7, 8].

In this work, we develop a geometric representation of two-dimensional spinors that overcomes these limitations. Starting from simple considerations about vectors, reference frames, and half-angle constructions, we build a visualization where:

- The *moduli* of the spinor components are encoded as the radii of two circle sections on a sphere;
- The *relative phase* between components appears as a geometric rotation of these circles;
- The *global phase* is also geometrically represented;
- The characteristic 4π -periodicity of spinors is made visually manifest.

The paper is organized as follows. In Section II we introduce a planar half-angle construction that serves as the foundation for the three-dimensional visualization. Section III promotes this construction to $\mathbb{R}^3$, defining the “hyperchord” circles that form our spinor representation. Section IV connects the geometric construction to the standard spinor formalism and shows how $SU(2)$ rotations act on the visualization. Section V presents the rendering algorithm with pseudocode. We conclude with a discussion in Section VI. An open-source Python implementation of the visualization, together with interactive Jupyter notebooks, is available at https://github.com/al-rigazzi/spinor_viz [9].

The reader interested in a comprehensive exposure of spinors and their algebra is referred to articles [10, 11] and monographs [5, 12, 13] on the subject.

II. PLANAR HALF-ANGLE CONSTRUCTION

Before constructing the full three-dimensional spinor visualization, we develop a simpler two-dimensional precursor that captures the essential half-angle geometry. This planar construction isolates the dependence of spinor *magnitudes* on a half-angle and motivates the spatial “hyperchord” construction used later.

A. Setup

Consider the unit circle $C = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$ centered at the origin. Fix the reference diameter along the x -axis, with endpoints

$$P_+ = (1, 0), \quad P_- = (-1, 0).$$

Let $Q(\vartheta) = (\cos \vartheta, \sin \vartheta)$ be a point on C , parametrized by ϑ .

B. The chord construction

For a given ϑ , define two chords:

1. the **upper chord** $\overline{P_+Q(\vartheta)}$,
2. the **lower chord** $\overline{P_-Q(\vartheta)}$.

A direct computation gives

$$|P_+Q(\vartheta)|^2 = (\cos \vartheta - 1)^2 + \sin^2 \vartheta = 2 - 2 \cos \vartheta = 4 \sin^2(\vartheta/2), \quad (1)$$

$$|P_-Q(\vartheta)|^2 = (\cos \vartheta + 1)^2 + \sin^2 \vartheta = 2 + 2 \cos \vartheta = 4 \cos^2(\vartheta/2), \quad (2)$$

hence

$$|P_+Q(\vartheta)| = 2 |\sin(\vartheta/2)|, \quad (3)$$

$$|P_-Q(\vartheta)| = 2 |\cos(\vartheta/2)|. \quad (4)$$

In particular,

$$|P_+Q(\vartheta)|^2 + |P_-Q(\vartheta)|^2 = 4,$$

which mirrors the identity $\sin^2(\vartheta/2) + \cos^2(\vartheta/2) = 1$.

a. Remark (choice of range). If one restricts to $\vartheta \in [0, \pi]$ (upper semicircle), then $\sin(\vartheta/2), \cos(\vartheta/2) \geq 0$ and the absolute values may be dropped. We keep ϑ on $[0, 2\pi)$ when discussing the 2π versus 4π behavior, but the planar chord *lengths* depend only on $|\sin(\vartheta/2)|$ and $|\cos(\vartheta/2)|$.

C. Connection to spinor components

For a normalized spinor, we adopt the symmetric phase convention

$$\xi = e^{i\gamma} \begin{bmatrix} e^{-i\varphi/2} \cos(\vartheta/2) \\ e^{+i\varphi/2} \sin(\vartheta/2) \end{bmatrix}, \quad \vartheta \in [0, \pi], \varphi \in [0, 2\pi), \gamma \in [0, 2\pi). \quad (5)$$

In this planar precursor we are concerned only with the *magnitudes*, so we set $\vartheta = \vartheta$ (and ignore φ, γ for the moment). Then

$$|\xi_{\uparrow}| = \cos(\vartheta/2), \quad (6)$$

$$|\xi_{\downarrow}| = \sin(\vartheta/2), \quad (7)$$

and, for $\vartheta \in [0, \pi]$,

$$|\xi_{\uparrow}| = |P_{-}Q(\vartheta)|/2, \quad (8)$$

$$|\xi_{\downarrow}| = |P_{+}Q(\vartheta)|/2. \quad (9)$$

Thus the chord lengths encode the spinor component magnitudes (up to a factor of 2).

D. Angle doubling and spinor periodicity

As ϑ ranges from 0 to 2π , the point $Q(\vartheta)$ traverses the circle once, whereas the half-angle $\vartheta/2$ ranges from 0 to π . Consequently, the *pair of chord lengths*—and therefore the pair of magnitudes $(|\xi_{\uparrow}|, |\xi_{\downarrow}|)$ —returns to its initial values after $\vartheta \mapsto \vartheta + 2\pi$.

However, in the spinorial representation of spatial rotations one has

$$U(\Theta) = \exp\left(-\frac{i}{2}\Theta \hat{n} \cdot \sigma\right) \in SU(2), \quad U(\Theta + 2\pi) = -U(\Theta),$$

so a continuous 2π rotation in physical space corresponds to a sign change of the spinor (and full return only after 4π). The planar chord *lengths* cannot encode this sign: they depend only on magnitudes and are therefore insensitive to an overall phase factor.

This motivates the spatial construction, where we augment the half-angle magnitude encoding with explicit phase/orientation markers so that ξ and $-\xi$ are visually distinct.

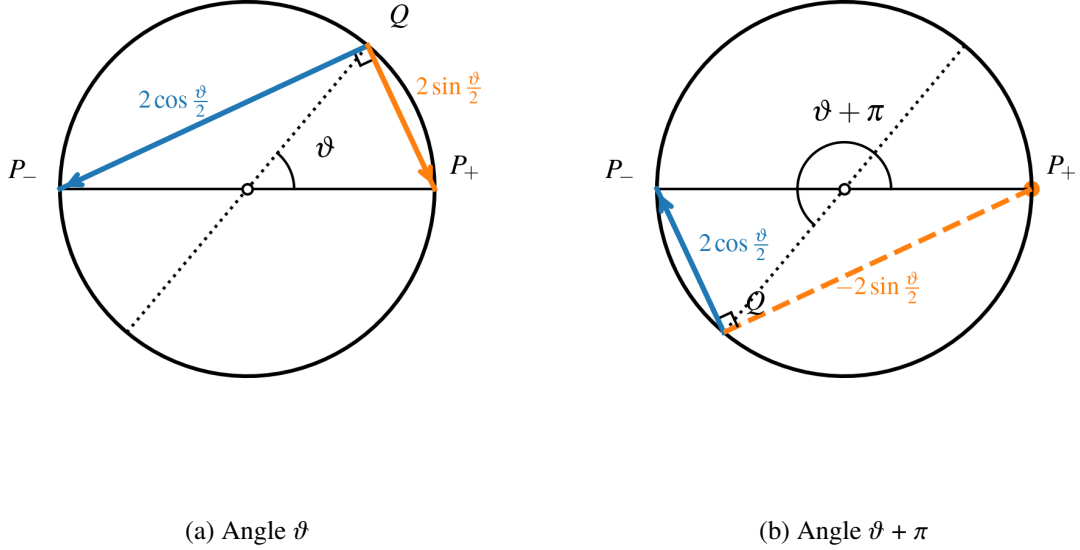


FIG. 1. Planar chord construction. (a) For angle ϑ , the chord from P_- to Q (blue) has length $2|\cos(\vartheta/2)|$, while the chord from P_+ to Q (red) has length $2|\sin(\vartheta/2)|$. The right angle at Q follows from Thales' theorem (triangle P_+QP_- subtends a diameter). (b) For $\vartheta + \pi$, the chord lengths exchange roles, reflecting the half-angle dependence.

E. From planar to spatial

The planar precursor encodes only the polar dependence (via half-angle magnitudes). To represent the full spinor data in (5), we must promote the construction to three dimensions:

- chord lengths become radii of plane–sphere intersection circles (the “hyperchord faces”);
- the *relative* phase φ is encoded as a relative orientation between the two faces;
- the fiber coordinate (global phase choice) γ is encoded as a common rotation of the phase markers.

This promotion is developed in Section III.

III. HYPERCHORD CONSTRUCTION IN THREE DIMENSIONS

We now promote the planar half-angle construction of Section II to three dimensions, obtaining the central geometric objects of this paper: two offset circle sections on a sphere, which we call

the *hyperchord faces*.

A. Setup: rotated equatorial plane and the highest equator point

a. Geometric scaling convention. A two-component $SU(2)$ spinor $\xi \in \mathbb{C}^2$ carries a natural quadratic invariant $u := \xi^\dagger \xi$. In this work we use u as the *geometric scale parameter*. Concretely, rather than embedding ξ directly, we embed the rescaled spinor

$$\tilde{\xi} := \|\xi\| \xi = \sqrt{u} \xi, \quad (10)$$

whose Euclidean norm is

$$\|\tilde{\xi}\| = \sqrt{\tilde{\xi}^\dagger \tilde{\xi}} = \sqrt{(\sqrt{u} \xi)^\dagger (\sqrt{u} \xi)} = \sqrt{u(\xi^\dagger \xi)} = u. \quad (11)$$

This is the reason the sphere used in the visualization has radius u . In dimensional terms, if one assigns spinors the conventional length-dimension $[\xi] = L^{1/2}$ (so that quadratic spinor bilinears carry dimension L), then $u = \xi^\dagger \xi$ has units of length and S_u is a genuine length-scale object. For normalized quantum states one has $u = 1$, and the construction reduces to the unit-radius case. For notational simplicity, we continue to write ξ for the input spinor and use $u := \xi^\dagger \xi$ to set the scene scale.

Fix the sphere of radius u ,

$$S_u := \{ \mathbf{x} \in \mathbb{R}^3 : \|\mathbf{x}\| = u \},$$

with $u := \xi^\dagger \xi$ as in (10)–(11). Note that the geometry of the two faces depends only on (ϑ, φ) , while u acts as an overall similarity scale and γ acts as a common in-face marker rotation.

and the reference equatorial plane $E := xy$ -plane with equator $e := E \cap S_u$. The primary datum is a rotated equatorial plane $E' = \mathbf{R}E$, for some $\mathbf{R} \in \text{SO}(3)$. Unprimed symbols refer to the reference configuration; primed symbols to the rotated one.

Let $\mathbf{v}$ be a chosen unit normal to E' (so E' determines $\mathbf{v}$ up to sign), and define

$$\vartheta \in [0, \pi] \quad \text{by} \quad \mathbf{v} \cdot \mathbf{e}_z = \cos \vartheta.$$

Equivalently, ϑ is the dihedral angle between E' and E (since $\mathbf{e}_z$ is the unit normal to E).

Let

$$e' := E' \cap S_u$$

denote the rotated equator. We single out a canonical point on e' by taking the maximizer of the z -coordinate:

$$Q := \arg \max_{\mathbf{x} \in e'} (\mathbf{x} \cdot \mathbf{e}_z).$$

For $\vartheta \in (0, \pi)$ this point is unique and can be written explicitly. Indeed, the unit vector in E' that maximizes $\mathbf{x} \cdot \mathbf{e}_z$ is the normalized projection of $\mathbf{e}_z$ onto E' , i.e.

$$\hat{\mathbf{h}} := \frac{\mathbf{e}_z - (\mathbf{e}_z \cdot \mathbf{v}) \mathbf{v}}{\|\mathbf{e}_z - (\mathbf{e}_z \cdot \mathbf{v}) \mathbf{v}\|} = \frac{\mathbf{e}_z - \cos \vartheta \mathbf{v}}{\sin \vartheta} \in E', \quad Q := u \hat{\mathbf{h}}.$$

In particular,

$$Q \cdot \mathbf{e}_z = u \sin \vartheta, \quad \text{and} \quad \mathbf{v} \cdot Q = 0.$$

(For the degenerate cases $\vartheta = 0$ or $\vartheta = \pi$, the entire equator e' lies in the xy -plane and the maximizer is not unique; in that case we fix Q by choosing an azimuth φ as discussed in Section III C.)

B. Definition of the two faces via dihedral bisectors

The hyperchord planes U' and L' are defined as follows.

First form the two unit *bisector normals*

$$\mathbf{n}_\uparrow := \frac{\mathbf{e}_z + \mathbf{v}}{\|\mathbf{e}_z + \mathbf{v}\|} = \frac{\mathbf{e}_z + \mathbf{v}}{2 \cos(\vartheta/2)}, \quad \mathbf{n}_\downarrow := \frac{\mathbf{e}_z - \mathbf{v}}{\|\mathbf{e}_z - \mathbf{v}\|} = \frac{\mathbf{e}_z - \mathbf{v}}{2 \sin(\vartheta/2)},$$

where the half-angle normalizations use $\|\mathbf{e}_z + \mathbf{v}\|^2 = 2 + 2 \cos \vartheta = 4 \cos^2(\vartheta/2)$ and $\|\mathbf{e}_z - \mathbf{v}\|^2 = 2 - 2 \cos \vartheta = 4 \sin^2(\vartheta/2)$. A direct computation gives

$$\mathbf{n}_\uparrow \cdot \mathbf{n}_\downarrow \propto (\mathbf{e}_z + \mathbf{v}) \cdot (\mathbf{e}_z - \mathbf{v}) = \mathbf{e}_z \cdot \mathbf{e}_z - \mathbf{v} \cdot \mathbf{v} = 0,$$

so the bisector normals are orthogonal.

Now define the *offset* planes as the unique translates of the corresponding bisector planes through the origin that pass through Q :

$$U' := \{ \mathbf{x} \in \mathbb{R}^3 : \mathbf{n}_\uparrow \cdot \mathbf{x} = \mathbf{n}_\uparrow \cdot Q \}, \quad L' := \{ \mathbf{x} \in \mathbb{R}^3 : \mathbf{n}_\downarrow \cdot \mathbf{x} = \mathbf{n}_\downarrow \cdot Q \}.$$

By construction $Q \in U' \cap L'$ and $U' \perp L'$ (since their normals are orthogonal).

The two hyperchord circles are then defined as the corresponding sphere sections

$$c_\uparrow := U' \cap S_u, \quad c_\downarrow := L' \cap S_u.$$

a. *Offsets and radii (half-angle law).* Because $Q \in E'$ we have $\mathbf{v} \cdot Q = 0$, hence

$$\mathbf{n}_\uparrow \cdot Q = \frac{(\mathbf{e}_z + \mathbf{v}) \cdot Q}{2 \cos(\vartheta/2)} = \frac{\mathbf{e}_z \cdot Q}{2 \cos(\vartheta/2)} = \frac{u \sin \vartheta}{2 \cos(\vartheta/2)} = u \sin(\vartheta/2),$$

and similarly

$$\mathbf{n}_\downarrow \cdot Q = \frac{(\mathbf{e}_z - \mathbf{v}) \cdot Q}{2 \sin(\vartheta/2)} = \frac{\mathbf{e}_z \cdot Q}{2 \sin(\vartheta/2)} = \frac{u \sin \vartheta}{2 \sin(\vartheta/2)} = u \cos(\vartheta/2).$$

Therefore the signed distances of the planes from the origin are

$$\text{dist}(U', 0) = u \sin(\vartheta/2), \quad \text{dist}(L', 0) = u \cos(\vartheta/2).$$

A plane at distance d from the origin cuts S_u in a circle of radius $\sqrt{u^2 - d^2}$, hence

$$r_\uparrow = \sqrt{u^2 - u^2 \sin^2(\vartheta/2)} = u \cos(\vartheta/2), \quad r_\downarrow = \sqrt{u^2 - u^2 \cos^2(\vartheta/2)} = u \sin(\vartheta/2),$$

which is precisely the desired half-angle scaling.

C. Connection to the equator and definition of the azimuth φ

Each hyperchord circle is tangent to the reference equator e at a single point, and these two tangent points are antipodal.

To see this, intersect U' with the xy -plane E (i.e. set $z = 0$). Using $\mathbf{n}_\uparrow = (\mathbf{e}_z + \mathbf{v})/(2 \cos(\vartheta/2))$ and $\mathbf{n}_\uparrow \cdot Q = u \sin(\vartheta/2)$, the plane equation $\mathbf{n}_\uparrow \cdot \mathbf{x} = \mathbf{n}_\uparrow \cdot Q$ reduces on E to

$$\mathbf{v} \cdot \mathbf{x} = u \sin \vartheta \quad (z = 0).$$

Writing $\mathbf{v} = (v_x, v_y, v_z)$ with $v_z = \cos \vartheta$ and $\sqrt{v_x^2 + v_y^2} = \sin \vartheta$, this is a line in E at distance

$$\frac{u \sin \vartheta}{\sqrt{v_x^2 + v_y^2}} = \frac{u \sin \vartheta}{\sin \vartheta} = u$$

from the origin. Hence the line $\ell_\uparrow := U' \cap E$ is tangent to the equator circle $e = \{(x, y, 0) : x^2 + y^2 = u^2\}$ at a *unique* point, which we denote by $P_\uparrow$. Explicitly, let $\widehat{\mathbf{v}}_{xy}$ be the unit projection of $\mathbf{v}$ onto E ,

$$\widehat{\mathbf{v}}_{xy} := \frac{1}{\sin \vartheta} (v_x, v_y, 0),$$

then the tangency point is

$$P_\uparrow := u \widehat{\mathbf{v}}_{xy} \in e.$$

The same calculation for L' gives a tangent line $\ell_{\downarrow} := L' \cap E$ whose unique tangency point is the antipode

$$P_{\downarrow} := -P_{\uparrow}.$$

We define the azimuthal angle φ by writing

$$P_{\uparrow} = u(\cos \varphi, \sin \varphi, 0), \quad P_{\downarrow} = u(\cos(\varphi + \pi), \sin(\varphi + \pi), 0).$$

In the visualization developed later, this φ is the geometric carrier of the *relative phase*: rotating the entire configuration about the z -axis rigidly shifts φ by the same amount, while the half-angle structure of the faces will induce the characteristic spinorial response.

D. Geometric meaning of the phases

Once the spinor is written in the symmetric half-angle phase split used throughout this work, the two faces carry distinguished in-face diameter directions whose *relative* rotation encodes the relative phase φ , while their *common* rotation encodes the fiber (global) phase γ . (We make this explicit after introducing the $SU(2)$ action on spinors in Section IV, using the symmetric phase split $\psi_{\uparrow} = \gamma - \varphi/2$, $\psi_{\downarrow} = \gamma + \varphi/2$.)

IV. CONNECTION TO $SU(2)$ AND THE BLOCH VECTOR

The hyperchord construction of Section III realizes, in geometric terms, the standard parametrization of a two-component spinor and the associated Bloch vector map. Throughout this section we keep the scale convention

$$u := \xi^{\dagger} \xi,$$

and we recall the geometric embedding convention introduced in Section III A: the rendered scene is scaled by the rescaled spinor

$$\tilde{\xi} := \|\xi\| \xi = \sqrt{u} \xi, \quad \text{so that} \quad \|\tilde{\xi}\| = u.$$

This is why the visualization sphere is $S_u = \{\mathbf{x} \in \mathbb{R}^3 : \|\mathbf{x}\| = u\}$. All direction-only quantities (e.g. the Bloch vector) are invariant under this scaling.

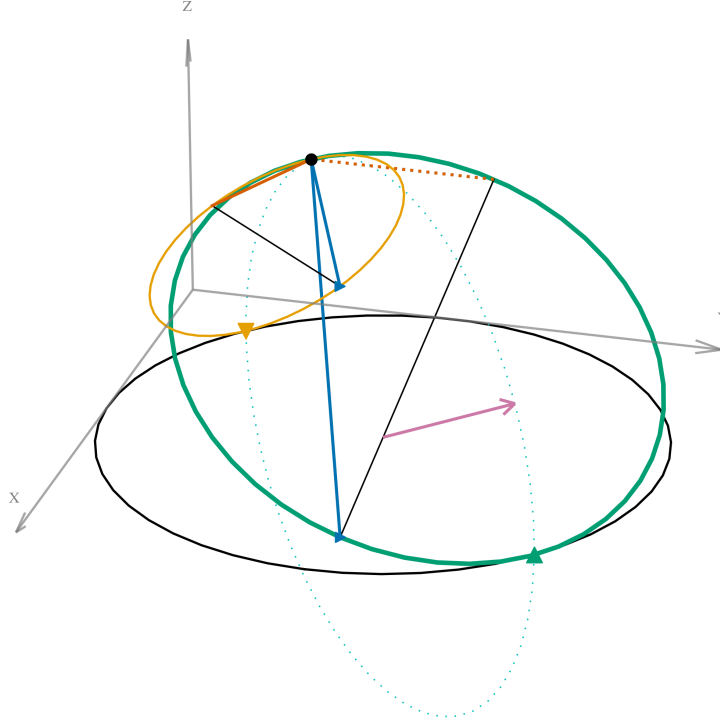


FIG. 2. Hyperchord construction. The rotated equatorial plane E' (primary datum) fixes the unit normal $\mathbf{v}$ and the dihedral angle ϑ with the reference plane $E = xy$ -plane. The rotated equator $e' = E' \cap S_u$ has a distinguished highest point Q . The two hyperchord planes U' and L' are the unique translates (through Q) of the two dihedral-bisector planes between E and E' , with normals proportional to $\mathbf{e}_z \pm \mathbf{v}$. Their sphere sections are the circles $c_\uparrow$ (teal/bluish-green, thicker line, equator contact marked with $\blacktriangle$) and $c_\downarrow$ (orange, thinner line, equator contact marked with $\blacktriangledown$), with radii $r_\uparrow = u \cos(\vartheta/2)$ and $r_\downarrow = u \sin(\vartheta/2)$. Each circle is tangent to the reference equator e at a unique point: $P_\uparrow$ for $c_\uparrow$ and $P_\downarrow = -P_\uparrow$ for $c_\downarrow$. The azimuth of $P_\uparrow$ is φ .

A. Standard half-angle parametrization

Let $\xi \in \mathbb{C}^2 \setminus \{0\}$ and write $\rho := \|\xi\| = \sqrt{u}$. Using the symmetric (Hopf-fiber) phase split adopted throughout this work, ξ can be written as

$$\xi = \rho e^{i\gamma} \begin{bmatrix} e^{-i\varphi/2} \cos(\vartheta/2) \\ e^{+i\varphi/2} \sin(\vartheta/2) \end{bmatrix}, \quad \vartheta \in [0, \pi], \quad \varphi \in [0, 2\pi), \quad \gamma \in [0, 2\pi). \quad (12)$$

Equivalently, in terms of component arguments

$$\xi_{\uparrow} = |\xi_{\uparrow}| e^{i\psi_{\uparrow}}, \quad \xi_{\downarrow} = |\xi_{\downarrow}| e^{i\psi_{\downarrow}},$$

one has

$$\varphi = \psi_{\downarrow} - \psi_{\uparrow} \pmod{2\pi}, \quad \gamma = \frac{1}{2}(\psi_{\uparrow} + \psi_{\downarrow}) \pmod{2\pi}, \quad \psi_{\uparrow} = \gamma - \varphi/2, \quad \psi_{\downarrow} = \gamma + \varphi/2. \quad (13)$$

The moduli are

$$|\xi_{\uparrow}| = \rho \cos(\vartheta/2), \quad |\xi_{\downarrow}| = \rho \sin(\vartheta/2), \quad \frac{|\xi_{\uparrow}|^2}{u} = \cos^2(\vartheta/2), \quad \frac{|\xi_{\downarrow}|^2}{u} = \sin^2(\vartheta/2).$$

a. Gauge-fixed form (common in the literature). A frequently used gauge chooses a representative in which the upper component is real and nonnegative, yielding

$$\xi = \rho e^{i\alpha} \begin{bmatrix} \cos(\vartheta/2) \\ e^{i\varphi} \sin(\vartheta/2) \end{bmatrix} \quad \text{for some } \alpha \in [0, 2\pi),$$

which is equivalent to (12) after absorbing a phase into α . We use the symmetric split (12) because it separates the relative phase φ from the common (fiber) phase γ in a way that matches the face-marker geometry.

B. Bloch vector and the SU(2)→SO(3) map

Let $\sigma_x, \sigma_y, \sigma_z$ denote the Pauli matrices. The (unit) Bloch vector associated with ξ is defined by

$$\mathbf{n}(\xi) := \frac{1}{\xi^\dagger \xi} \begin{bmatrix} \xi^\dagger \sigma_x \xi \\ \xi^\dagger \sigma_y \xi \\ \xi^\dagger \sigma_z \xi \end{bmatrix} \in \mathbb{R}^3, \quad \|\mathbf{n}(\xi)\| = 1, \quad (14)$$

and for the half-angle parametrization (12) one obtains the standard expression

$$\mathbf{n}(\xi) = \begin{bmatrix} \sin \vartheta \cos \varphi \\ \sin \vartheta \sin \varphi \\ \cos \vartheta \end{bmatrix}. \quad (15)$$

This is the usual SU(2) double-cover map to SO(3) (via the adjoint action on Pauli matrices); see [5, 14].

In the present visualization, the *primary datum* is the rotated equatorial plane E' (Section III A), with chosen unit normal $\mathbf{v}$. We identify this normal with the Bloch direction,

$$\mathbf{v} = \mathbf{n}(\boldsymbol{\xi}), \quad (16)$$

so that the polar angle ϑ appearing in Section III A (defined by $\mathbf{v} \cdot \mathbf{e}_z = \cos \vartheta$) is exactly the Bloch polar angle in (15). (Choosing the opposite normal $-\mathbf{v}$ corresponds to $\vartheta \mapsto \pi - \vartheta$ together with the exchange of the two components; the construction is otherwise unchanged.)

The azimuth φ is read geometrically from the equator via the tangency points $P_\uparrow$ and $P_\downarrow$ (Section III C).

C. Born-rule weights from face areas

From Section III B, the two hyperchord circles have radii

$$r_\uparrow = u \cos(\vartheta/2), \quad r_\downarrow = u \sin(\vartheta/2).$$

Hence the corresponding disk areas

$$A_\uparrow = \pi r_\uparrow^2 = \pi u^2 \cos^2(\vartheta/2), \quad A_\downarrow = \pi r_\downarrow^2 = \pi u^2 \sin^2(\vartheta/2)$$

satisfy the normalized relations

$$\frac{A_\uparrow}{\pi u^2} = \cos^2(\vartheta/2) = \frac{\|\boldsymbol{\xi}_\uparrow\|^2}{u}, \quad \frac{A_\downarrow}{\pi u^2} = \sin^2(\vartheta/2) = \frac{\|\boldsymbol{\xi}_\downarrow\|^2}{u}. \quad (17)$$

Thus, at fixed u , the hyperchord face areas encode the standard two-level Born weights.

D. Equivalent encoding via minimal projected lengths

A second (independent) encoding comes from *minimal* orthogonal projections onto the xy -plane. The clean way to see it is by slicing the sphere with vertical planes parallel to the meridian plane determined by the common point Q and the tangency points $P_\uparrow, P_\downarrow$ (cf. Figure 2).

Fix such a vertical slice at signed distance $|y_d| \leq u$ from the origin. Its intersection with the sphere is a circle of radius

$$r_d = \sqrt{u^2 - y_d^2}.$$

In that slice, the two relevant equator points are $(\pm r_d, 0)$ (in (x, z) -coordinates), while the highest point of the rotated equator occurs at

$$H'(y_d) = (-r_d \cos \vartheta, r_d \sin \vartheta),$$

(where, without loss of generality, we have rotated azimuth so that the meridian direction coincides with the x -axis; the final formulas are azimuth-independent).

Now consider the two Euclidean segments in the slice connecting $H'(y_d)$ to the two equator points $(\pm r_d, 0)$. Their orthogonal projections onto the xy -plane coincide with their x -projections in the slice, and therefore have lengths

$$\ell_+(y_d) = r_d(1 + \cos \vartheta), \quad \ell_-(y_d) = r_d(1 - \cos \vartheta). \quad (18)$$

At $y_d = 0$ this yields the extremal projected chord lengths

$$\ell_+(0) = u(1 + \cos \vartheta) = 2u \cos^2(\vartheta/2), \quad \ell_-(0) = u(1 - \cos \vartheta) = 2u \sin^2(\vartheta/2),$$

so that the normalized ratio reproduces the Born weights:

$$\frac{\ell_+(0)}{\ell_+(0) + \ell_-(0)} = \cos^2(\vartheta/2), \quad \frac{\ell_-(0)}{\ell_+(0) + \ell_-(0)} = \sin^2(\vartheta/2). \quad (19)$$

E. Equivalent encoding via projected areas (slicing proof)

Integrating the slice widths (18) over all vertical slices gives the projected areas on the xy -plane of the two corresponding surface patches (equivalently: the projected areas obtained by stacking the slice projections):

$$\Pi_+ = \int_{-u}^u \ell_+(y_d) dy_d = (1 + \cos \vartheta) \int_{-u}^u \sqrt{u^2 - y_d^2} dy_d = (1 + \cos \vartheta) \frac{\pi u^2}{2} = \pi u^2 \cos^2(\vartheta/2), \quad (20)$$

$$\Pi_- = \int_{-u}^u \ell_-(y_d) dy_d = (1 - \cos \vartheta) \frac{\pi u^2}{2} = \pi u^2 \sin^2(\vartheta/2). \quad (21)$$

Hence

$$\frac{\Pi_+}{\Pi_+ + \Pi_-} = \cos^2(\vartheta/2), \quad \frac{\Pi_-}{\Pi_+ + \Pi_-} = \sin^2(\vartheta/2), \quad (22)$$

showing that *disk areas*, *minimal projected lengths*, and *projected slice-integrated areas* all encode the same Born weights.

F. Geometric meaning of φ and γ

Finally, the remaining angular degrees of freedom in (12) are visible as in-face rotations of the two hyperchord faces. Using (13), the relative phase φ is the difference of component arguments, $\varphi = \psi_{\downarrow} - \psi_{\uparrow}$, while the fiber (global) phase γ is their average, $\gamma = \frac{1}{2}(\psi_{\uparrow} + \psi_{\downarrow})$.

As summarized in Section III D, the distinguished diameter directions drawn on $c_{\uparrow}$ and $c_{\downarrow}$ rotate within their respective face planes by angles determined by $\psi_{\uparrow}$ and $\psi_{\downarrow}$. Therefore:

- the *relative* twist between the two face markers encodes φ ,
- a *common* twist of both markers encodes γ (i.e. the choice of representative along the $U(1)$ Hopf fiber).

In contrast, bilinear quantities such as the Bloch vector $\mathbf{n}(\xi)$ are insensitive to γ and depend only on (ϑ, φ) .

V. VISUALIZATION ALGORITHM

This section specifies a concrete mapping

$$\xi = \begin{bmatrix} \xi_{\uparrow} \\ \xi_{\downarrow} \end{bmatrix} \in \mathbb{C}^2 \quad \mapsto \quad (S_u, c_{\uparrow}, c_{\downarrow}, \text{phase markers})$$

suitable for implementation in a 3D renderer. All geometric objects are expressed in the fixed Cartesian frame $\{\mathbf{e}_x, \mathbf{e}_y, \mathbf{e}_z\}$ and use the paper's convention

$$S_u := \{\mathbf{x} \in \mathbb{R}^3 : \|\mathbf{x}\| = u\}, \quad u := \xi^{\dagger} \xi = |\xi_{\uparrow}|^2 + |\xi_{\downarrow}|^2.$$

(Thus u sets the sphere radius; the spinor's Euclidean norm is $\sqrt{u}$. In the common normalized case $u = 1$.)

A. Overview

Given ξ , the visualization proceeds in three stages:

1. **Extract** $(u, \vartheta, \varphi, \gamma)$ from the spinor components, where ϑ controls magnitudes, φ is the relative phase, and γ is a chosen Hopf-fiber (common) phase.

2. **Construct** the two hyperchord circles $c_\uparrow$ and $c_\downarrow$ as explicit plane–sphere intersections, with radii matching $u \cos(\vartheta/2)$ and $u \sin(\vartheta/2)$.
3. **Encode phases** by drawing oriented diameter markers on each circle, determined by the component arguments $\arg(\xi_\uparrow)$ and $\arg(\xi_\downarrow)$.

B. Parameter extraction

Let

$$u := |\xi_\uparrow|^2 + |\xi_\downarrow|^2, \quad \rho := \sqrt{u}. \quad (23)$$

Define the half-angle moduli

$$\cos \frac{\vartheta}{2} := \frac{|\xi_\uparrow|}{\rho}, \quad \sin \frac{\vartheta}{2} := \frac{|\xi_\downarrow|}{\rho}, \quad \text{so that} \quad \vartheta = 2 \arctan 2(|\xi_\downarrow|, |\xi_\uparrow|) \in [0, \pi]. \quad (24)$$

Write the component phases as

$$\xi_\uparrow = |\xi_\uparrow| e^{i\psi_\uparrow}, \quad \xi_\downarrow = |\xi_\downarrow| e^{i\psi_\downarrow}, \quad \psi_\uparrow := \arg(\xi_\uparrow), \quad \psi_\downarrow := \arg(\xi_\downarrow) \pmod{2\pi}.$$

Define the relative phase and a symmetric (fiber) phase by

$$\varphi := \psi_\downarrow - \psi_\uparrow \pmod{2\pi}, \quad \gamma := \frac{1}{2}(\psi_\uparrow + \psi_\downarrow) \pmod{2\pi}. \quad (25)$$

With these definitions one has the standard symmetric spinor form

$$\xi = \rho e^{i\gamma} \begin{bmatrix} e^{-i\varphi/2} \cos \frac{\vartheta}{2} \\ e^{+i\varphi/2} \sin \frac{\vartheta}{2} \end{bmatrix}. \quad (26)$$

(Equivalently, fixing a gauge by setting $\gamma = \varphi/2$ reduces (26) to the familiar form $\rho e^{i\alpha} [\cos \frac{\vartheta}{2}, e^{i\varphi} \sin \frac{\vartheta}{2}]^T$ with upper component real and nonnegative.)

a. Degenerate cases. If $|\xi_\uparrow| = 0$ or $|\xi_\downarrow| = 0$, the corresponding argument is undefined and the relative phase φ is not encoded in the geometry. In such cases one may set $\varphi := 0$ and choose γ from the nonzero component (e.g. $\gamma := \psi_\downarrow$ when $|\xi_\uparrow| = 0$).

C. Circle construction from closed-form plane data

Introduce the radial and azimuthal unit vectors in the equatorial plane,

$$\mathbf{e}_r(\varphi) := \cos \varphi \mathbf{e}_x + \sin \varphi \mathbf{e}_y, \quad \mathbf{e}_\varphi(\varphi) := -\sin \varphi \mathbf{e}_x + \cos \varphi \mathbf{e}_y, \quad (27)$$

so that $\{\mathbf{e}_r(\varphi), \mathbf{e}_\varphi(\varphi), \mathbf{e}_z\}$ is right-handed.

a. *Equator contact points.* Define the two antipodal equator points

$$P_{\uparrow} := u \mathbf{e}_r(\varphi) \in e, \quad P_{\downarrow} := -u \mathbf{e}_r(\varphi) \in e, \quad (28)$$

which are the tangency points of $c_{\uparrow}$ and $c_{\downarrow}$ with the equator in the reference view.

b. *Hyperchord plane normals and offsets.* Define the unit normals of the two *offset* planes U' and L' by

$$\hat{\mathbf{n}}_{\uparrow} := \sin \frac{\vartheta}{2} \mathbf{e}_r(\varphi) + \cos \frac{\vartheta}{2} \mathbf{e}_z, \quad \hat{\mathbf{n}}_{\downarrow} := \cos \frac{\vartheta}{2} \mathbf{e}_r(\varphi) - \sin \frac{\vartheta}{2} \mathbf{e}_z. \quad (29)$$

These satisfy $\hat{\mathbf{n}}_{\uparrow} \cdot \hat{\mathbf{n}}_{\downarrow} = 0$, hence $U' \perp L'$. The signed plane equations are chosen so that $P_{\uparrow} \in U'$ and $P_{\downarrow} \in L'$:

$$U' : \hat{\mathbf{n}}_{\uparrow} \cdot \mathbf{x} = d_{\uparrow}, \quad L' : \hat{\mathbf{n}}_{\downarrow} \cdot \mathbf{x} = -d_{\downarrow}, \quad (30)$$

where

$$d_{\uparrow} := \hat{\mathbf{n}}_{\uparrow} \cdot P_{\uparrow} = u \sin \frac{\vartheta}{2}, \quad d_{\downarrow} := -(\hat{\mathbf{n}}_{\downarrow} \cdot P_{\downarrow}) = u \cos \frac{\vartheta}{2}. \quad (31)$$

(These are exactly the offsets stated in Section III B.)

c. *Circle centers and radii.* For a unit-normal plane $\hat{\mathbf{n}} \cdot \mathbf{x} = d$, the circle $S_u \cap \{\hat{\mathbf{n}} \cdot \mathbf{x} = d\}$ has center $d\hat{\mathbf{n}}$ and radius $\sqrt{u^2 - d^2}$. Thus,

$$C_{\uparrow} = d_{\uparrow} \hat{\mathbf{n}}_{\uparrow}, \quad r_{\uparrow} = \sqrt{u^2 - d_{\uparrow}^2} = u \cos \frac{\vartheta}{2}, \quad (32)$$

$$C_{\downarrow} = (-d_{\downarrow}) \hat{\mathbf{n}}_{\downarrow}, \quad r_{\downarrow} = \sqrt{u^2 - d_{\downarrow}^2} = u \sin \frac{\vartheta}{2}. \quad (33)$$

Equivalently (using $\rho = \sqrt{u}$),

$$r_{\uparrow} = \rho |\xi_{\uparrow}|, \quad r_{\downarrow} = \rho |\xi_{\downarrow}|. \quad (34)$$

d. *Shared point (useful for validation).* The two planes (30) intersect the sphere in circles that share the point

$$Q := u(-\cos \vartheta \mathbf{e}_r(\varphi) + \sin \vartheta \mathbf{e}_z) \in S_u, \quad (35)$$

which is the highest point of the rotated equator $e' = E' \cap S_u$ (cf. Section III A). One checks directly that $Q \in U' \cap L'$ by dotting with $\hat{\mathbf{n}}_{\uparrow}$ and $\hat{\mathbf{n}}_{\downarrow}$.

D. Parametric rendering of the inclined circles

To sample points on each circle, choose an orthonormal basis in each plane. A convenient choice is to reuse the azimuthal direction $\mathbf{e}_{\varphi}(\varphi)$, which is orthogonal to both $\mathbf{e}_r(\varphi)$ and $\mathbf{e}_z$, hence lies in both planes:

$$\mathbf{b} := \mathbf{e}_{\varphi}(\varphi), \quad \mathbf{a}_{\uparrow} := \mathbf{b} \times \hat{\mathbf{n}}_{\uparrow}, \quad \mathbf{a}_{\downarrow} := \mathbf{b} \times \hat{\mathbf{n}}_{\downarrow}. \quad (36)$$

Then $\{\mathbf{a}_\uparrow, \mathbf{b}\}$ is an orthonormal basis of U' , and $\{\mathbf{a}_\downarrow, \mathbf{b}\}$ is an orthonormal basis of L' . The circle parameterizations are

$$\mathbf{x}_\uparrow(t) = C_\uparrow + r_\uparrow(\cos t \mathbf{a}_\uparrow + \sin t \mathbf{b}), \quad t \in [0, 2\pi), \quad (37)$$

$$\mathbf{x}_\downarrow(t) = C_\downarrow + r_\downarrow(\cos t \mathbf{a}_\downarrow + \sin t \mathbf{b}), \quad t \in [0, 2\pi). \quad (38)$$

A renderer can discretize $t_k = 2\pi k/N$ and draw the resulting polyline.

E. Phase markers

Each complex component defines an in-plane direction (a “flag” on the corresponding face). Using $\psi_\uparrow$ and $\psi_\downarrow$ from above, define the two diameter directions

$$\mathbf{d}_\uparrow := \cos \psi_\uparrow \mathbf{a}_\uparrow + \sin \psi_\uparrow \mathbf{b}, \quad \mathbf{d}_\downarrow := \cos \psi_\downarrow \mathbf{a}_\downarrow + \sin \psi_\downarrow \mathbf{b}, \quad (39)$$

and render (for example) the oriented segments

$$\text{upper marker: } C_\uparrow \rightarrow C_\uparrow + r_\uparrow \mathbf{d}_\uparrow, \quad \text{lower marker: } C_\downarrow \rightarrow C_\downarrow + r_\downarrow \mathbf{d}_\downarrow. \quad (40)$$

The *relative* phase φ is visible as the relative rotation between the two markers, since $\varphi = \psi_\downarrow - \psi_\uparrow$. A common shift $\psi_\uparrow \mapsto \psi_\uparrow + \delta$ and $\psi_\downarrow \mapsto \psi_\downarrow + \delta$ rotates both markers by the same amount in their respective faces; in terms of (25) this is precisely $\gamma \mapsto \gamma + \delta$ at fixed φ .

If one prefers to display real/imaginary axes explicitly, draw two orthogonal arrows on each face along $\mathbf{d}$ and $\mathbf{d}^\perp$ with $\mathbf{d}^\perp := (-\sin \psi)\mathbf{a} + (\cos \psi)\mathbf{b}$, optionally scaling them by $|\Re(\xi_i)|$ and $|\Im(\xi_i)|$ and encoding the sign by line style.

F. Complete scene assembly

A minimal visualization consists of:

1. the sphere S_u (or just its equator e);
2. the circles $c_\uparrow$ and $c_\downarrow$ via (37)–(38), drawn with distinct colors and line widths for visual discriminability (see Appendix A for the specific encoding);
3. the equator contact points $P_\uparrow$ and $P_\downarrow$, marked with upward ($\blacktriangle$) and downward ($\blacktriangledown$) triangles, respectively;

4. the phase markers (40).

Optional elements include the Bloch vector (computed from the normalized spinor) and a meridian arc through $P_{\uparrow}$ and $P_{\downarrow}$ to guide the eye.

VI. DISCUSSION AND CONCLUSIONS

A. Comparison with existing visualizations

The hyperchord representation offers several advantages over existing spinor visualization methods:

a. Bloch sphere. The standard Bloch sphere represents a spinor by a single point on the unit sphere, corresponding to the Bloch vector $\mathbf{v}$. This representation:

- ✓ Elegantly captures the direction (ratio of components);
- × Loses global phase information entirely;
- × Does not distinguish spinors that differ by a sign (ξ vs. $-\xi$);
- × Obscures the 4π -periodicity.

The hyperchord representation retains all spinorial degrees of freedom while still displaying the Bloch vector as the normal to the rotated equatorial plane.

b. Stereographic projections. Stereographic representations map the spinor to points in the complex plane. While mathematically elegant, they:

- × Break the visual symmetry between components;
- × Do not display the sphere geometry;
- × Require familiarity with projective geometry.

c. Arrow/flag constructions. Some authors visualize spinors using oriented objects (arrows with flags, or twisted ribbons) that acquire a twist under 2π rotations. Our approach is similar in spirit but more systematic: the hyperchord circles and their phase markers provide a complete, easily parsable representation.

B. Physical interpretation

The hyperchord construction has a natural interpretation in terms of quantum measurement and state geometry:

- The *face areas* encode the Born-rule weights for measuring spin-up vs. spin-down along the z -axis (equivalently: the squared moduli of the computational-basis amplitudes), via $A_{\uparrow}/(\pi u^2) = |\xi_{\uparrow}|^2/u$ and $A_{\downarrow}/(\pi u^2) = |\xi_{\downarrow}|^2/u$.
- The *relative phase* $\varphi = \arg(\xi_{\downarrow}) - \arg(\xi_{\uparrow})$ controls the orientation of the Bloch vector in the equatorial plane and therefore affects measurement statistics along axes other than z (e.g. along x or y).
- The remaining phase parameter $\gamma = \frac{1}{2}(\arg(\xi_{\uparrow}) + \arg(\xi_{\downarrow}))$ is a *fiber (global) phase*: it is a choice of representative along the $U(1)$ Hopf fiber and does not affect single-qubit measurement probabilities. It becomes operationally relevant only when the state is compared to a phase reference or another branch/subsystem—for example in interferometric settings where relative phases between paths are measured, or when describing composite systems where only relative phases between components of a superposition have physical meaning.

C. Two-dimensional projection

While the three-dimensional hyperchord representation provides complete geometric information, it can be challenging to interpret in static printed figures where interactive exploration is not possible. For such cases, we provide a two-dimensional projection onto the equatorial (xy) plane.

The 2D projection preserves all essential spinorial information:

- The **equator** projects to itself (a circle of radius u);
- The **equator contact points** mark where each hyperchord meets the equator, encoding the azimuthal phase φ . The upper hyperchord's contact point $P_{\uparrow}$ is marked with an upward-pointing triangle ($\blacktriangle$) in teal/bluish-green, and the lower hyperchord's contact point $P_{\downarrow}$ with a downward-pointing triangle ($\blacktriangledown$) in orange;
- The **hyperchord circles** are visually distinguished by both color and line thickness: $c_{\uparrow}$ is drawn with a thicker line in teal/bluish-green and $c_{\downarrow}$ with a thinner line in orange;

- The **circle radii** $r_{\uparrow} = u \cos(\vartheta/2)$ and $r_{\downarrow} = u \sin(\vartheta/2)$ are preserved, directly encoding the Born-rule probabilities via their areas;
- The **phase arrows** (blue for real part, vermillion for imaginary part) are drawn on each circle with solid lines and arrowheads for positive values and dashed lines with filled circles for negative values.

Two rendering modes are available:

1. **Circle mode** (default): The hyperchord faces are drawn as circles with their true 3D radii. This provides the clearest visualization of the probability amplitudes, as the disk areas directly encode the Born-rule probabilities.
2. **Ellipse mode**: The hyperchord faces are drawn as ellipses, representing their true orthogonal projection onto the xy -plane. The major axis equals the hyperchord radius, while the minor axis is scaled by $\cos(\beta)$ where β is the tilt angle of the hyperchord plane.

Figure 3 illustrates both modes for a general spinor state. In circle mode, the two circles touch at the projection of their shared 3D contact point, preserving the tangency condition while displaying the true hyperchord radii. In ellipse mode, the projected ellipses show the geometric distortion from the inclined 3D planes.

The circle mode is recommended for most applications, as it preserves the intuitive area-probability correspondence while avoiding the visual distortion of elliptical projections.

D. Animation and dynamics

The hyperchord visualization is particularly effective for animating spinor dynamics. The reference implementation (Section V and Appendix A), provided as the open-source Python package `spinor_viz` [9], supports frame-by-frame animation of:

- Rotations about arbitrary axes, showing how the hyperchord circles smoothly transform;
- The 4π periodicity, by animating a full 4π rotation and observing the return to initial configuration;
- Precession dynamics, relevant for magnetic resonance applications.

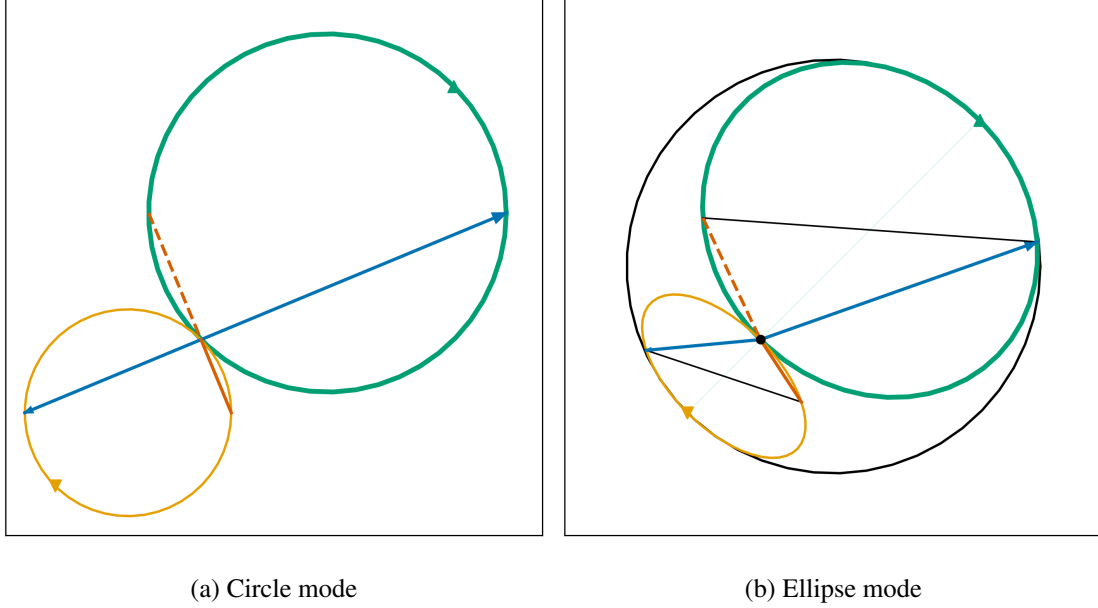


FIG. 3. Two-dimensional projection of the hyperchord visualization for the spinor $\xi = [\cos(\pi/6), \sin(\pi/6)e^{i\pi/4}]^T$. (a) Circle mode: the hyperchord faces are drawn as tangent circles with their true 3D radii. (b) Ellipse mode: the hyperchord faces are drawn as their orthogonal projections onto the xy -plane.

Figure 4 illustrates the evolution of a spinor under rotation about the z -axis. The initial state has $\vartheta = \pi/3$ and $\varphi = \pi/4$. As the rotation angle increases from 0 to 2π , the hyperchord circles rotate about their respective axes, demonstrating the half-angle relationship: a 2π physical rotation corresponds to the spinor acquiring a phase of π , visually manifested as a π rotation of the hyperchord orientations.

The same rotation sequence is shown in the 2D projection modes in Figures 5 and 6.

E. Extensions

The hyperchord construction naturally extends to:

- **Four-component spinors:** The $SL(2, \mathbb{C})$ generalization uses hyperchords in $\mathbb{R}^{3+1}$ to visualize Dirac spinors (to be presented in a companion paper);
- **Mixed states:** Density matrices can be represented by “fuzzy” hyperchords with reduced radii reflecting purity;

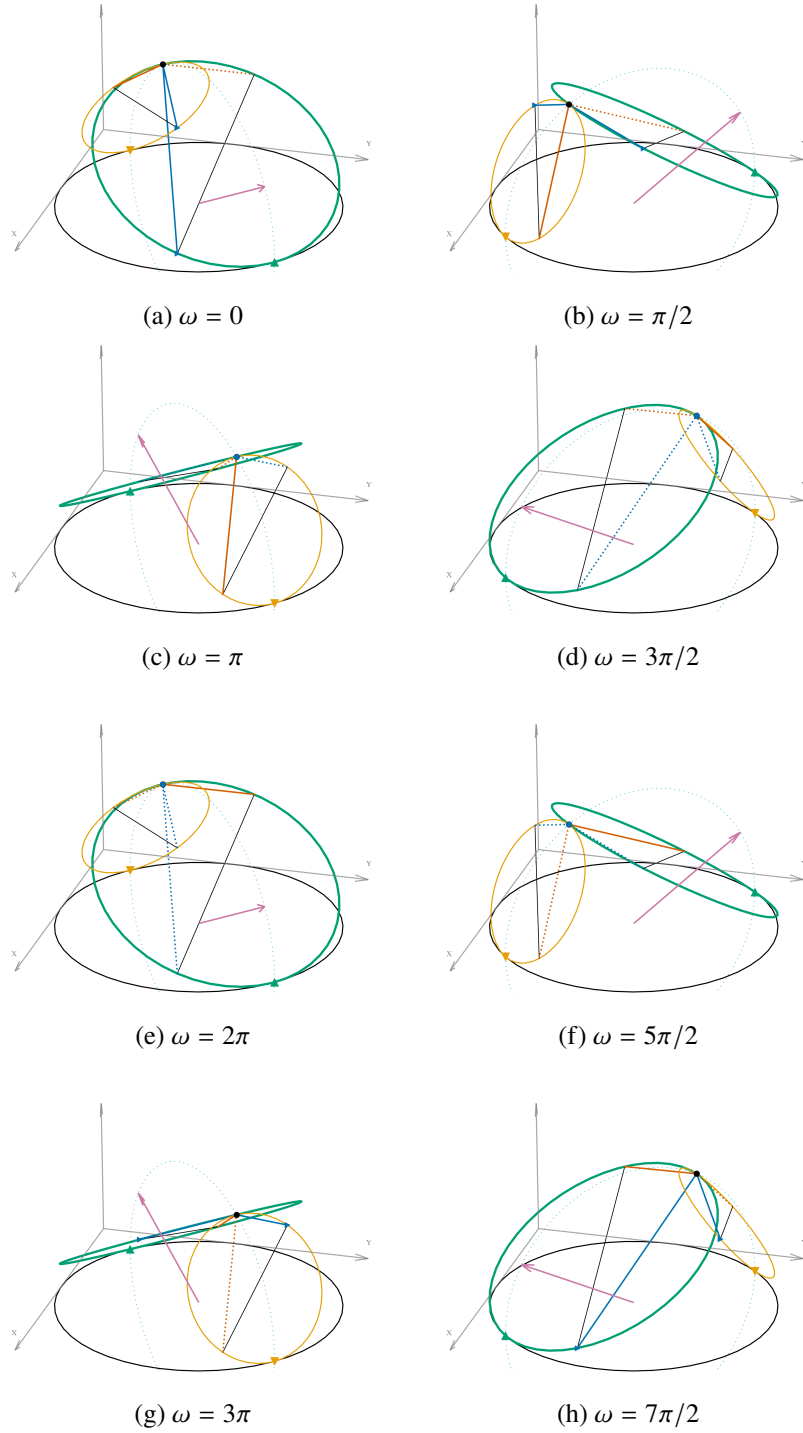


FIG. 4. Animation sequence of a spinor undergoing rotation about the z -axis. Starting from $\vartheta = \pi/3$, $\varphi = \pi/4$, the spinor is rotated by angle ω via the operator $e^{-i\omega\sigma_z/2}$. After a 2π rotation (frame e), the spinor has returned to the same Bloch vector but acquired a π phase shift, visible in the hyperchord orientations. Full return to the initial state requires a 4π rotation.

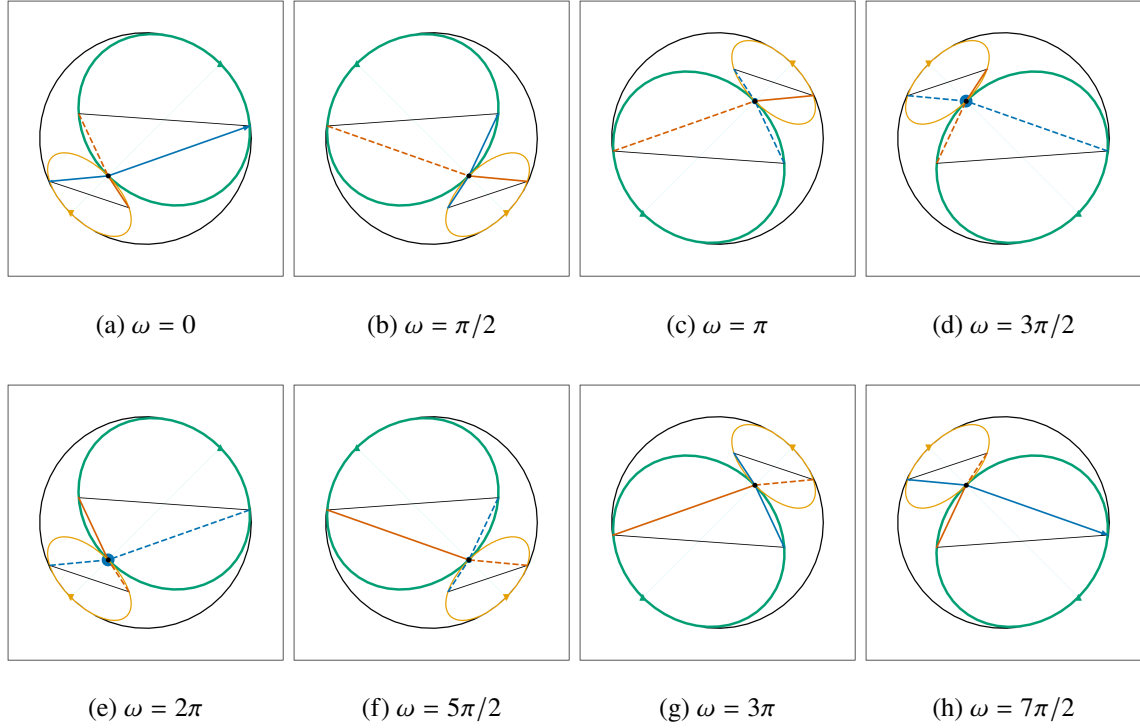


FIG. 5. 2D ellipse-mode projection of the z -rotation sequence from Fig. 4. The projected ellipses show the true orthogonal projection of the inclined hyperchord circles onto the xy -plane.

- **Multi-qubit systems:** Tensor product structure can be visualized using nested or linked hyperchord diagrams.

F. Conclusions

We have presented a novel geometric construction for visualizing two-dimensional spinors. The hyperchord representation:

1. Faithfully encodes all spinorial degrees of freedom (moduli and phases);
2. Makes the half-angle structure geometrically transparent;
3. Displays the 4π -periodicity characteristic of spinors;
4. Connects naturally to the Bloch sphere while providing additional information;
5. Admits a straightforward rendering algorithm suitable for static figures and animations.

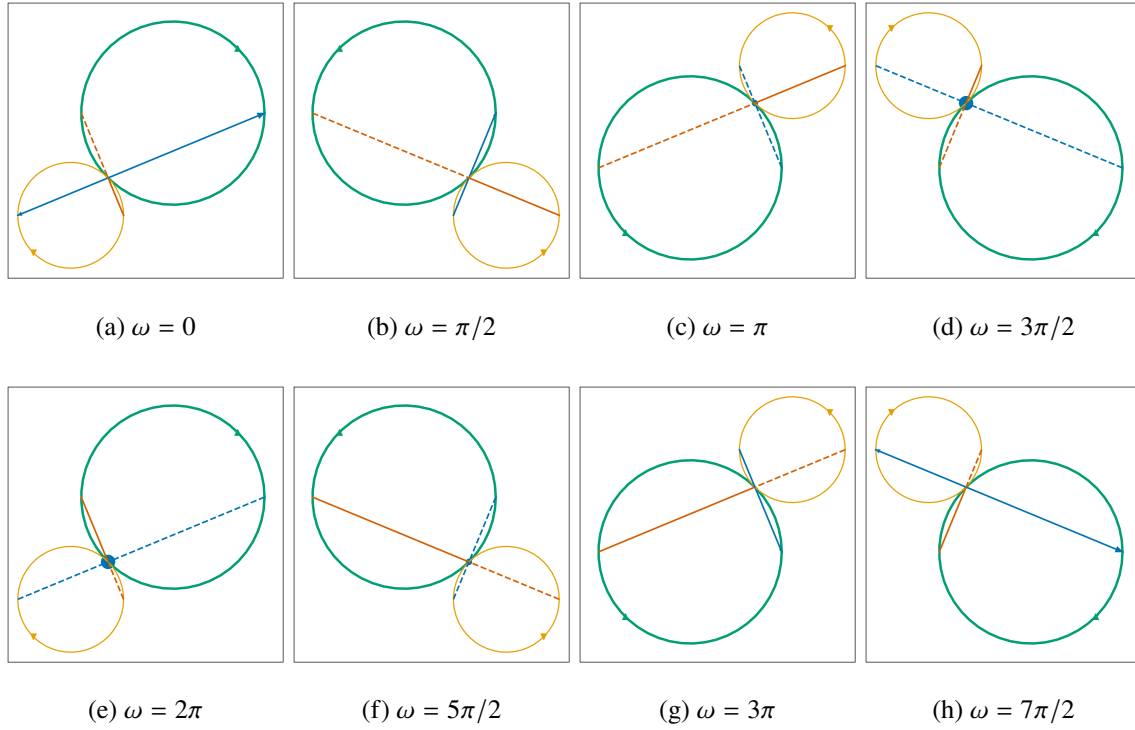


FIG. 6. 2D circle-mode projection of the z -rotation sequence from Fig. 4. The tangent circles preserve the true hyperchord radii, with their areas encoding the Born-rule probabilities.

By grounding the visualization in the simple half-angle chord construction of Section II, we provide an intuitive geometric foundation that illuminates why spinors behave as they do. The construction serves both pedagogical and practical purposes, offering researchers and students a new tool for understanding and communicating spinorial physics.

A complete open-source Python implementation of the visualization algorithm is available at https://github.com/al-rigazzi/spinor_viz [9]. The repository includes the `spinor_viz` library, which provides both static Matplotlib and interactive Plotly visualizations, as well as a set of Jupyter notebooks that allow readers to interactively explore spinor states, rotations, quantum gate actions, and animations without writing any code.

Appendix A: Visualization Algorithm Details

This appendix provides complete pseudocode for the spinor visualization algorithm, consistent with the conventions of Section V. The pseudocode follows the structure of the reference Python implementation (the `spinor_viz` package), which uses the symmetric phase parametrization

adopted in the paper.

a. Implementation note. The reference Python implementation, available as an open-source package at https://github.com/al-rigazzi/spinor_viz, uses a colorblind-friendly palette (the Wong palette [15]) and distinct visual encodings to distinguish the two hyperchord circles. Specifically:

- The *upper hyperchord* $c_{\uparrow}$ is drawn with a thicker line (linewidth 3.0) in teal/bluish-green (#009E73), and its kissing point with the equator $P_{\uparrow}$ is marked with an upward-pointing triangle ($\blacktriangle$).
- The *lower hyperchord* $c_{\downarrow}$ is drawn with a thinner line (linewidth 1.5) in orange (#E69F00), and its kissing point with the equator $P_{\downarrow}$ is marked with a downward-pointing triangle ($\blacktriangledown$).
- Phase-marker arrows use blue (#0072B2) for the real part and vermillion (#D55E00) for the imaginary part, with solid lines and arrowheads for positive values and dashed lines with filled circles for negative values.
- The Bloch vector is drawn in reddish-purple (#CC79A7).

These color and thickness choices ensure that the two hyperchord circles remain distinguishable both in color and in achromatic reproductions.

1. Main visualization routine

The main routine takes a spinor $\xi \in \mathbb{C}^2$ and renders the complete hyperchord visualization.

Procedure: PlotSpinor3D

Input: Spinor $\xi = [\xi_{\uparrow}, \xi_{\downarrow}]^T$, view direction (optional)

Output: 3D plot of hyperchord visualization

1. *Extract parameters and phases:*

- $(\vartheta, \varphi, \gamma, \psi_{\uparrow}, \psi_{\downarrow}, u, \rho) \leftarrow \text{SpinorToAngles}(\xi)$
- (Here $u = \xi^{\dagger} \xi$ and $\rho = \sqrt{u} = \|\xi\|$.)

2. *Define equatorial unit vectors:*

- $\mathbf{e}_r \leftarrow [\cos \varphi, \sin \varphi, 0]$

- $\mathbf{e}_\varphi \leftarrow [-\sin \varphi, \cos \varphi, 0]$

3. *Compute equator contact points:*

- $P_\uparrow \leftarrow u \mathbf{e}_r$
- $P_\downarrow \leftarrow -u \mathbf{e}_r$

4. *Compute half-angle moduli (for readability):*

- $c \leftarrow \cos(\vartheta/2) = |\xi_\uparrow|/\rho$
- $s \leftarrow \sin(\vartheta/2) = |\xi_\downarrow|/\rho$

5. *Compute plane normals (Section V, Eq. (29)):*

- $\hat{\mathbf{n}}_\uparrow \leftarrow s \mathbf{e}_r + c \mathbf{e}_z$
- $\hat{\mathbf{n}}_\downarrow \leftarrow c \mathbf{e}_r - s \mathbf{e}_z$

6. *Compute plane offsets (Eq. (30)):*

- $d_\uparrow \leftarrow \hat{\mathbf{n}}_\uparrow \cdot P_\uparrow = u s$
- $d_\downarrow \leftarrow -(\hat{\mathbf{n}}_\downarrow \cdot P_\downarrow) = u c$

7. *Compute circle centers and radii:*

- $C_\uparrow \leftarrow d_\uparrow \hat{\mathbf{n}}_\uparrow$
- $C_\downarrow \leftarrow (-d_\downarrow) \hat{\mathbf{n}}_\downarrow$
- $r_\uparrow \leftarrow \sqrt{u^2 - d_\uparrow^2} = u c$
- $r_\downarrow \leftarrow \sqrt{u^2 - d_\downarrow^2} = u s$

8. *Compute shared point (optional validation; Eq. (35)):*

- $Q \leftarrow u(-\cos \vartheta \mathbf{e}_r + \sin \vartheta \mathbf{e}_z)$

9. *Draw reference elements:*

- Draw equator e as the circle of radius u in the xy -plane.
- Optionally draw the sphere S_u as a transparent surface.

10. Choose in-plane bases for parameterization:

- $\mathbf{b} \leftarrow \mathbf{e}_\varphi$
- $\mathbf{a}_\uparrow \leftarrow \mathbf{b} \times \hat{\mathbf{n}}_\uparrow$ (normalize if needed)
- $\mathbf{a}_\downarrow \leftarrow \mathbf{b} \times \hat{\mathbf{n}}_\downarrow$ (normalize if needed)

11. Draw upper hyperchord circle if $r_\uparrow > \epsilon$:

- Parameterize circle points $\mathbf{x}_\uparrow(t) = C_\uparrow + r_\uparrow(\cos t \mathbf{a}_\uparrow + \sin t \mathbf{b})$ for $t \in [0, 2\pi)$.
- Draw $c_\uparrow$ in teal/bluish-green (thicker line).
- Mark equator contact point $P_\uparrow$ with an upward triangle ($\blacktriangle$).
- Draw the upper phase marker (Step 11).

12. Draw lower hyperchord circle if $r_\downarrow > \epsilon$:

- Parameterize circle points $\mathbf{x}_\downarrow(t) = C_\downarrow + r_\downarrow(\cos t \mathbf{a}_\downarrow + \sin t \mathbf{b})$ for $t \in [0, 2\pi)$.
- Draw $c_\downarrow$ in orange (thinner line).
- Mark equator contact point $P_\downarrow$ with a downward triangle ($\blacktriangledown$).
- Draw the lower phase marker (Step 11).

13. Draw phase markers (diameter directions):

- $\mathbf{d}_\uparrow \leftarrow \cos \psi_\uparrow \mathbf{a}_\uparrow + \sin \psi_\uparrow \mathbf{b}$
- $\mathbf{d}_\downarrow \leftarrow \cos \psi_\downarrow \mathbf{a}_\downarrow + \sin \psi_\downarrow \mathbf{b}$
- Draw oriented segments:

$$C_\uparrow \rightarrow C_\uparrow + r_\uparrow \mathbf{d}_\uparrow, \quad C_\downarrow \rightarrow C_\downarrow + r_\downarrow \mathbf{d}_\downarrow.$$

(The relative rotation between the markers encodes $\varphi = \psi_\downarrow - \psi_\uparrow$, while the common shift of both ψ 's corresponds to $\gamma \mapsto \gamma + \delta$ at fixed φ .)

14. Draw Bloch vector (optional):

- $\mathbf{v} \leftarrow \text{SpinorToBloch}(\xi)$
- Draw arrow from origin to $\mathbf{v}$ (or to $u \mathbf{v}$ depending on your scene scaling).

2. Spinor to angles conversion

Procedure: SpinorToAngles

Input: Spinor $\xi = [\xi_{\uparrow}, \xi_{\downarrow}]^T$

Output: $(\vartheta, \varphi, \gamma, \psi_{\uparrow}, \psi_{\downarrow}, u, \rho)$

1. $u \leftarrow |\xi_{\uparrow}|^2 + |\xi_{\downarrow}|^2, \quad \rho \leftarrow \sqrt{u}$
2. $\vartheta/2 \leftarrow \arctan 2(|\xi_{\downarrow}|, |\xi_{\uparrow}|), \quad \vartheta \leftarrow 2(\vartheta/2)$
3. If $|\xi_{\uparrow}| > 0$, set $\psi_{\uparrow} \leftarrow \arg(\xi_{\uparrow})$; else set $\psi_{\uparrow} \leftarrow 0$
4. If $|\xi_{\downarrow}| > 0$, set $\psi_{\downarrow} \leftarrow \arg(\xi_{\downarrow})$; else set $\psi_{\downarrow} \leftarrow 0$
5. $\varphi \leftarrow \psi_{\downarrow} - \psi_{\uparrow} \pmod{2\pi}$
6. $\gamma \leftarrow \frac{1}{2}(\psi_{\uparrow} + \psi_{\downarrow}) \pmod{2\pi}$
7. Return $(\vartheta, \varphi, \gamma, \psi_{\uparrow}, \psi_{\downarrow}, u, \rho)$

3. Spinor rotation

Procedure: SpRot

Input: Axis-angle vector $\omega = \omega \hat{\mathbf{n}}$

Output: 2×2 unitary rotation matrix $\mathbf{S}$

1. Compute rotation magnitude: $\omega \leftarrow \|\omega\|$
2. Compute unit axis: $\hat{\mathbf{n}} \leftarrow \omega/\omega$
3. Compute generator: $\mathbf{M} \leftarrow n_x \sigma_x + n_y \sigma_y + n_z \sigma_z$
4. Compute rotation matrix:

$$\mathbf{S} \leftarrow \exp(-i \mathbf{M} \omega/2) \quad (\text{equivalently } \mathbf{S} = \cos(\omega/2)\mathbf{I} - i \sin(\omega/2)\mathbf{M}).$$

5. Return $\mathbf{S}$

4. Spinor to Bloch vector

Procedure: SpinorToBloch

Input: Spinor ξ

Output: Bloch vector $\mathbf{v} \in \mathbb{R}^3$

1. $v_x \leftarrow \xi^\dagger \sigma_x \xi / (\xi^\dagger \xi)$
2. $v_y \leftarrow \xi^\dagger \sigma_y \xi / (\xi^\dagger \xi)$
3. $v_z \leftarrow \xi^\dagger \sigma_z \xi / (\xi^\dagger \xi)$
4. Return $\mathbf{v} = [v_x, v_y, v_z]$

5. Animation routine

Procedure: AnimateSpinor

Input: Initial spinor ξ_0 , rotation axis $\hat{\mathbf{n}}$, number of frames N

Output: Animation frames

1. Set $\beta_{\max} \leftarrow 4\pi$ for a full 4π rotation (to show spinor periodicity)
2. For $k = 1$ to N :
 - $\beta \leftarrow \beta_{\max} \cdot (k - 1) / (N - 1)$
 - $\mathbf{S} \leftarrow \text{SpRot}(\hat{\mathbf{n}} \cdot \beta)$
 - $\xi \leftarrow \mathbf{S} \cdot \xi_0$
 - Clear frame, draw rotation axis, call $\text{PlotSpinor3D}(\xi)$
 - Capture frame

6. Pauli matrices

For reference, the Pauli matrices used throughout:

$$\sigma_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \sigma_y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \quad \sigma_z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}. \quad (\text{A1})$$

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